netquest

Visualize It!

Introduction to Data Visualization

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1. Introduction

Introduction
Data visualization

The ways we structure and visualize information are changing rapidly and getting more complex with each passing day. **Thanks to the rise of social media, the ubiquity of mobile devices, and service digitalization, data is available on any human activity that utilizes technology**. The generated information is hugely valuable and makes it possible to analyze trends and patterns, and to use big data to draw connections between events. Thus, data visualization can be an effective mechanism for presenting the end user with understandable information in real time.

Every company has data, be it to communicate with clients and senior managers or to help manage the organization itself. It is only through research and interpretation that this data can acquire meaning and be transformed into knowledge.

This ebook seeks to guide readers through a series of basic references in order to help them understand data visualization and its component parts, and to equip them with the tools and platforms they need to create interactive visuals and analyze data. In effect, it seeks to provide readers with a basic vocabulary and a crash course in the principles of design that govern data visualization so that they can create and analyze interactive market research reports.

What is data visualization?

Data visualization is the process of acquiring, interpreting and comparing data in order to clearly communicate complex ideas, thereby facilitating the identification and analysis of meaningful patterns.

> Data visualization can be essential to strategic communication: it helps us interpret available data; detect patterns, trends, and anomalies; make decisions; and analyze inherent processes.
> All told, it can have a powerful impact on the business world.

The data visualization process

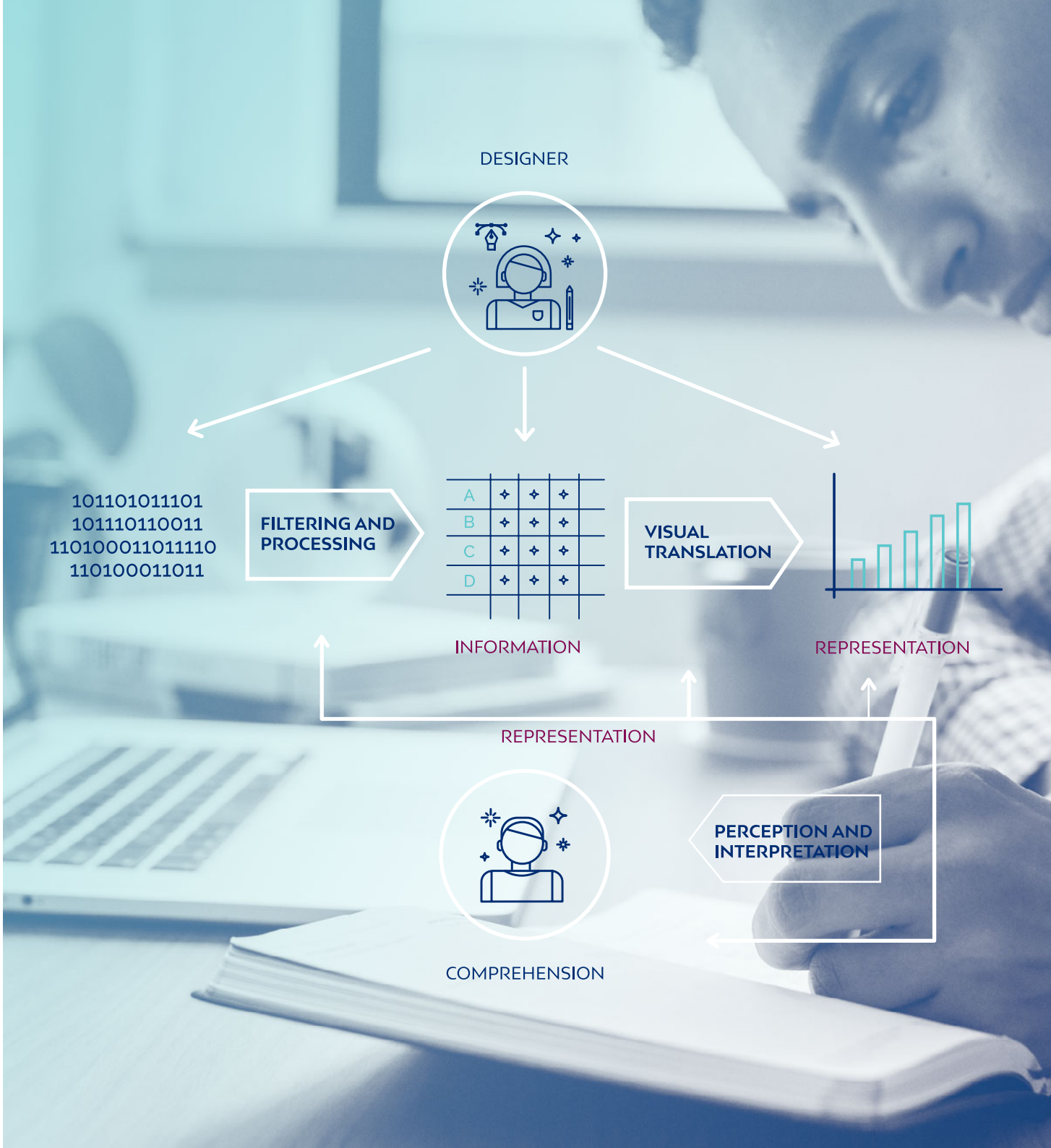
Several different fields are involved in the data visualization process, with the aim of simplifying or revealing existing relationships, or discovering something new within a data set.

Visualization process[1]

Filtering & processing. Refining and cleaning data to convert it into information through analysis, interpretation, contextualization, comparison, and research.

Translation & visual representation. Shaping the visual representation by defining graphic resources, language, context, and the tone of the representation, all of which are adapted for the recipient.

Perception & interpretation. Finally, the visualization becomes effective when it has a perceptive impact on the construction of knowledge.



1 Pérez, J. and Vialcanet, G. (2013). Guía de visualización de datos aplicada al marketing digital: Cómo transformar datos en conocimiento (p.5-6).

Why is data visualization so important in reports and statements?

We live in the era of visual information, and visual content plays an important role in every moment of our lives. A study by SH!FT Disruptive Learning demonstrated that **we typically process images 60,000 times faster than a table or a text,** and that our brains typically do a better job remembering them in the long term. That same research detected that after three days, analyzed subjects retained between 10% and 20% of written or spoken information, compared with 65% of visual information.

The rationale behind the power of visuals:

- The human mind can see an image for just **13 milliseconds** and store the information, provided that it is associated with a concept. Our eyes can take in **36,000 visual messages per hour**.
- **40%** of nerve fibers are connected to the retina.

All of this indicates that human beings are better at processing visual information, which is lodged in our long-term memory.

Consequently, for reports and statements, a visual representation that uses images is a much more effective way to communicate information than text or a table; it also takes up much less space.

This means that **data visuals are more attractive, simpler to take in, and easier to remember.**

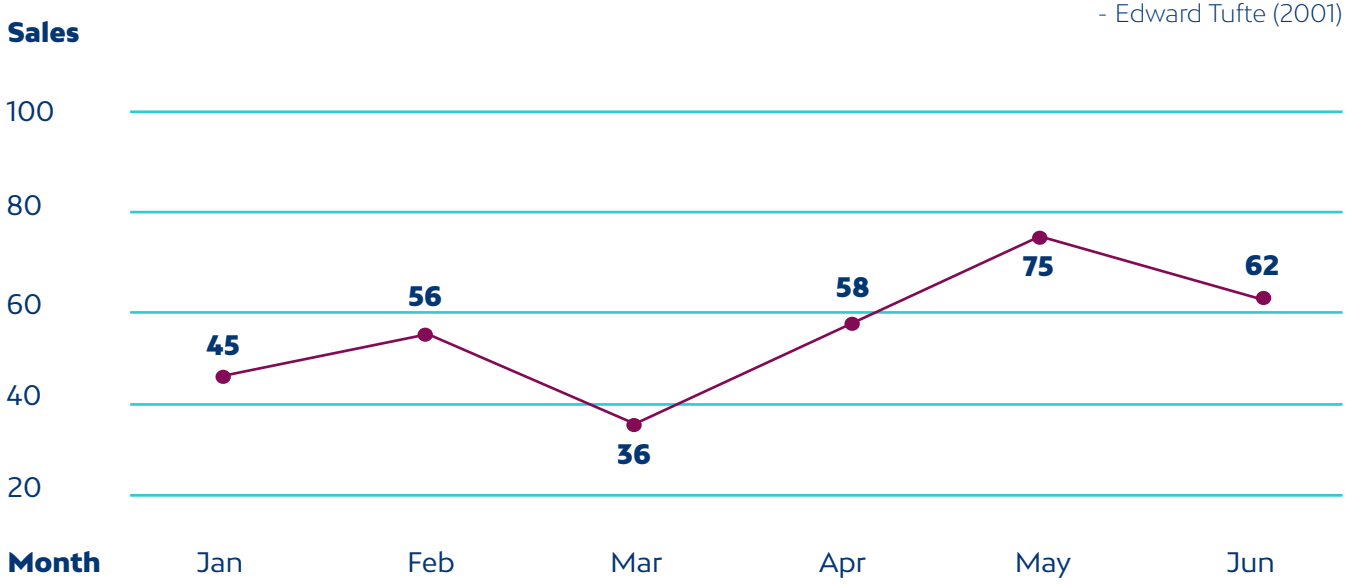
Try it for yourself. Take a look at this table:

Month	Jan	Feb	Mar	Apr	May	Jun
Sales	45	56	36	58	75	62

Identifying the evolution of sales over the course of the year isn't easy. However, when we present the same information in a visual, the results are much clearer (see the graph below).

The graph takes what the numbers cannot communicate on their own and conveys it in a visible, memorable way. This is the real strength of data visualization.

> **Graphical excellence is that which gives to the viewer the greatest number of ideas in the shortest time with the least ink in the smallest space."**
>
> - Edward Tufte (2001)

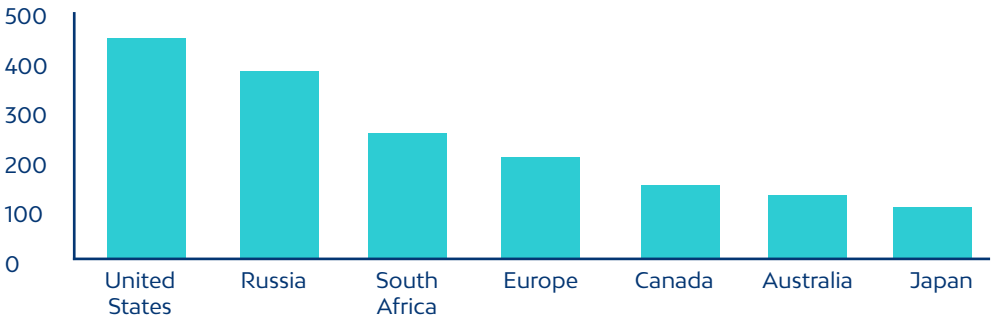


Data visualization chiefly helps in 3 key aspects of reports and statements:

1) Explaining

Visuals aim to lead the viewer down a path in order to describe situations, answer questions, support decisions, communicate information, or solve specific problems. When you attempt to explain something through data visualization, you start with a question, which interacts with the data set in such a way that enables viewers to make a decision and, subsequently, answer the question.

For example: This graphic below could clearly explain the country with the greatest demand for a certain product compared globally, in a concrete month.



2) Exploring

Some visuals are designed to lend a data set spatial dimensions, or to offer numerous subsets of data in order to raise questions, find answers, and discover opportunities. When the goal of a visual is to explore, the viewers start by familiarizing themselves with the dataset, then identifying an area of interest, asking questions, exploring, and finding several solutions or answers.

For example: an interactive graphic from *The Guardian*[2] invites us to explore how the linguistic standard of U.S. presidential addresses has declined over time. The visual is interactive and explanatory, in addition to indicating the readability score of various presidents' speeches.

3) Analyzing

Other visuals prompt viewers to inspect, distill, and transform the most significant information in a data set so that they can discover something new or predict upcoming situations.

For example: this interactive graphic about learning machine[3] invites us to explore and discover information within the visual by scrolling through it. Using the machine learning method, the visual explains the patterns detected in the data in order to categorize characteristics.

We'll close this introduction with a 2012 reflection by Alberto Cairo, a specialist in information visualization and a leader in the world of data visualization. For the author, a good visual must provide clarity, highlight trends, uncover patterns, and reveal unseen realities:

We create visuals so that users can analyze data and, from it, discover realities that not even the designer, in some instances, had considered."

2 Available at: https://www.fusioncharts.com/whitepapers/downloads/Principles-of-Data-Visualization.pdf
3 Available at: http://www.r2d3.us/visual-intro-to-machine-learning-part-1/

2. Data types, relationships, and visualization formats

Data types, relationships, and visualization formats

There are a number of methods and approaches to creating visuals based on the nature and complexity of the data and the information. **Different kinds of graphics are used in data visualizations, including representations of statistics, maps, and diagrams.** These schematic, visual representations of content vary in their degree of abstraction.

In order to communicate effectively, it is important to understand different kinds of data and to establish visual relationships through the proper use of graphics. Enrique Rodríguez (2012), a data analyst at DataNauta, once explained in an interview that...

> **A good graphic is one that synthesizes and contextualizes all of the information that's necessary to understand a situation and decide how to move forward."**

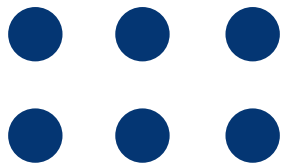
2 kinds of data

Before we talk about visuals themselves, we must first understand the different kinds of data that can be visualized and how they relate to one another.
The most common kinds of data are[4]:

1) Quantitative (numeric)

Data that can be quantified and measured. This kind of data explains a trend or the results of research through numeric values. This category of data can be further subdivided into:

- **Discrete:** Data that consists of whole numbers (0, 1, 2, 3...). For example, the number of children in a family.
- **Continuous:** Data that can take any value within an interval. For example, people's height (between 60 - 70 inches) or weight (between 90 and 110 pounds).

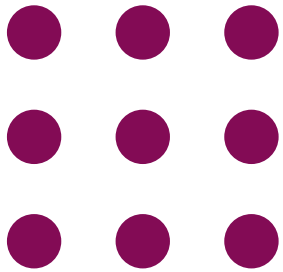


Quantitative

2) Qualitative (categoric)

This kind of data is divided into categories based on non-numeric characteristics. It may or may not have a logical order, and it measures qualities and generates categorical answers. It can be:

- **Ordinal:** Meaning it follows an order or sequence. That might be the alphabet or the months of the year.
- **Categorical:** Meaning it follows no fixed order. For example, varieties of products sold.



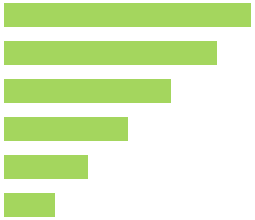
Qualitative

5 Source: Hubspot, Prezy, and Infogram (2018). Presenting Data People Can't Ignore: How to Communicate Effectively Using Data. | p.10 of 16 | Available at: https://offers.hubspot.com/presenting-data-people-cant-ignore.

7 data relationships

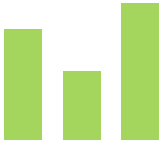
Data relationships can be simple, like the progress of a single metric over time (such as visits to a blog over the course of 30 days or the number of users on a social network), or they can be complex, precisely comparing relationships, revealing structure, and extracting patterns from data. There are **seven data relationships** to consider:

Ranking: A visualization that relates two or more values with respect to a relative magnitude. For example: a company's most sold products.



Deviation: Examines how each data point relates to the others and, particularly, to what point its value differs from the average. For example: the line of deviation for tickets to an amusement park sold on a rainy versus a normal day.

Nominal comparisons: Visualizations that compare quantitative values from different subcategories. For example: product prices in various supermarkets.



Correlation: Data with two or more variables that can demonstrate a positive or negative correlation with one another. For example: salaries based on level of education.

Partial and total relationships: Show a subset of data as compared with a larger total. For example: the percentage of clients that buy specific products.

Series over time: Here we can trace the changes in the values of a constant metric over the course of time. For example: monthly sales of a product over the course of two years.

Distribution: Visualization that shows the distribution of data spatially, often around a central value. For example: the heights of players on a basketball team.

11 formats

There are two types of visualizations: static and interactive. Their use depends on the search and analysis dimension level. **Static visuals can only analyze data in one dimension, whereas interactive visuals can analyze it in several.**

As with any other form of communication, familiarity with the code and resources that are available to us is essential if we're going to use them successfully our goal. In this page, we present the different kinds of graphics that we can use to transform our data into information. This group of visualization types is listed in order of popularity in the "Visualization Universe" project by Google News Lab and Adioma, as of the publication of this report.

1. Bar chart

Bar charts are one of the most popular ways of visualizing data because they present a data set in a quickly understood format that enables viewers to identify highs and lows at a glance.

They are very versatile, and they are typically used to compare discrete categories, to analyze changes over time, or to compare parts of a whole.
The three variations on the bar chart are:

Vertical column

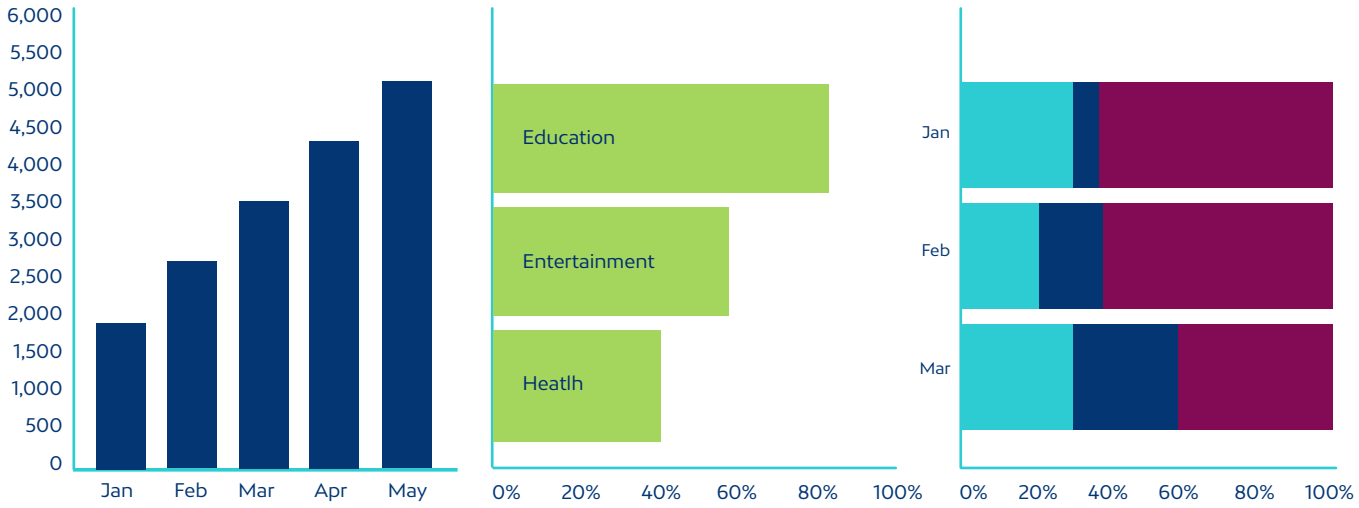
Used for chronological data, and it should be in left-to-right format.

Horizontal column

Used to visualize categories.

Full stacked column

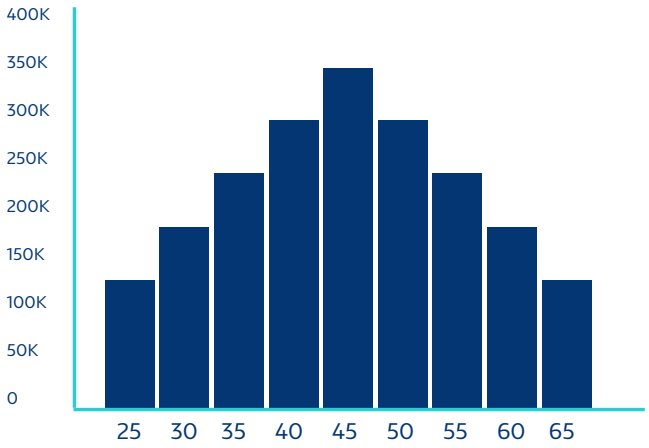
Used to visualize categories that collectively add up to 100%.



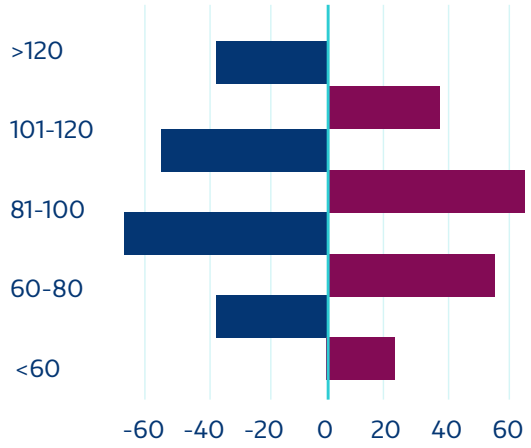
2. Histograms

Histograms represent a **variable in the form of bars, where the surface of each bar is proportional to the frequency of the values represented.** They offer an overview of the distribution of a population or sample with respect to a given characteristic.
The two variations on the histogram are:

- **Vertical columns**
- **Horizontal columns**



Vertical columns



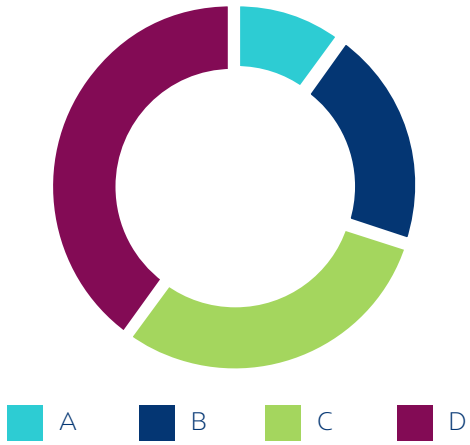
Horizontal columns

3. Pie charts

Pie charts consist of **a circle divided into sectors, each of which represents a portion of the total.** They can be subdivided into no more than five data groups. They can be useful for comparing discrete or continuous data.
The two variations on the pie chart are:

- **Standard:** Used to exhibit relationship between parts.
- **Donut:** A stylistic variation that facilitates the inclusion of a total value or a design element in the center.

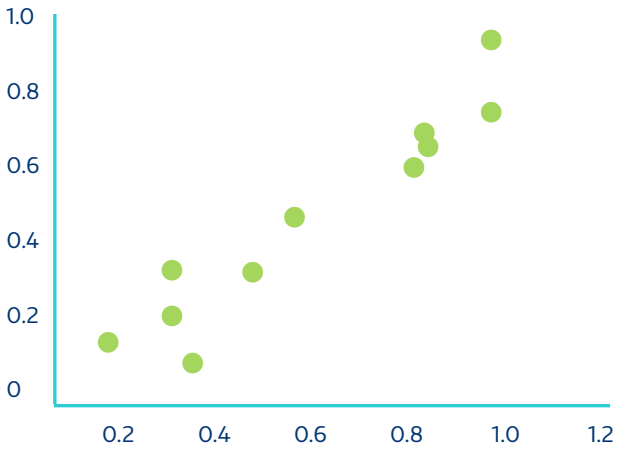
Standard pie chart



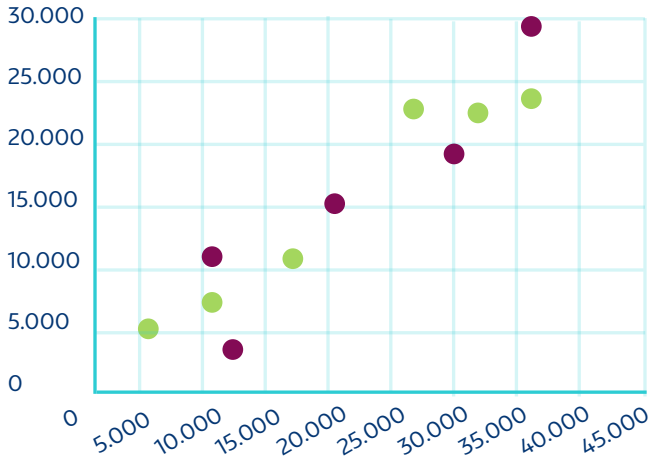
Donut pie chart

4. Scatter plots

Scatter plots use the spread of points over a Cartesian coordinate plane to show the relationship between two variables. They also help us determine whether or not different groups of data are correlated.



Scatter plot

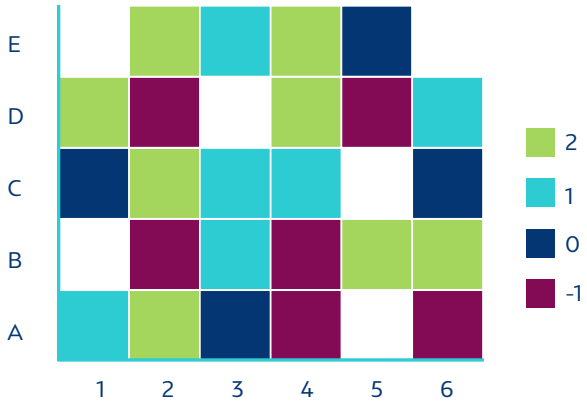


Scatter plot with grid

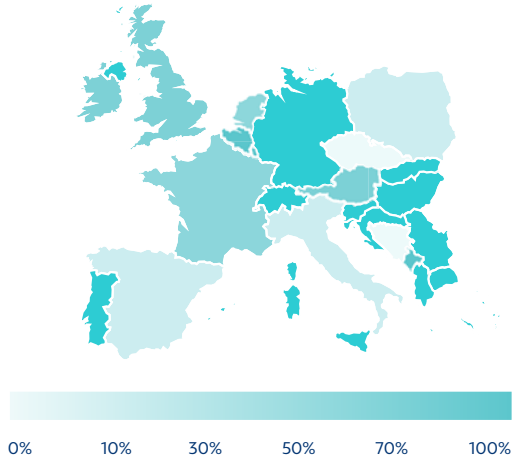
5. Heat maps

Heat maps represent individual values from a data set on a matrix using variations in color or color intensity. They often use color to help viewers compare and distinguish between data in two different categories at a glance. They are useful for visualizing webpages, where the areas that users interact with most are represented with "hot" colors, and the pages that receive the fewest clicks are presented in "cold" colors. The two variations on the heat map are:

- **Mosaic diagram**
- **Color map**



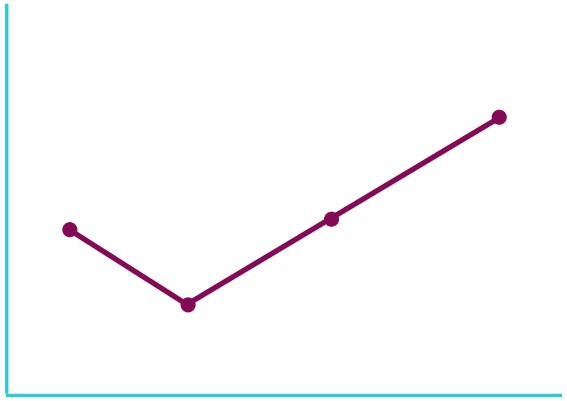
Mosaic diagram



Color map

6. Line charts

These are **used to display changes or trends in data over a period of time.** They are especially useful for showcasing relationships, acceleration, deceleration, and volatility in a data set.

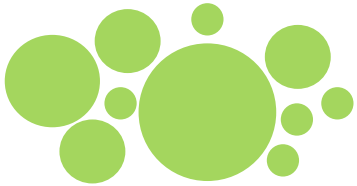


Line chart

7. Bubble charts

These graphics **display three-dimensional data and accentuate data in dispersion diagrams and maps.** Their purpose is to highlight nominal comparisons and classification relationships. The size and color of the bubbles represent a dimension that, along with the data, is very useful for visually stressing specific values. The two variations on the bubble chart are:

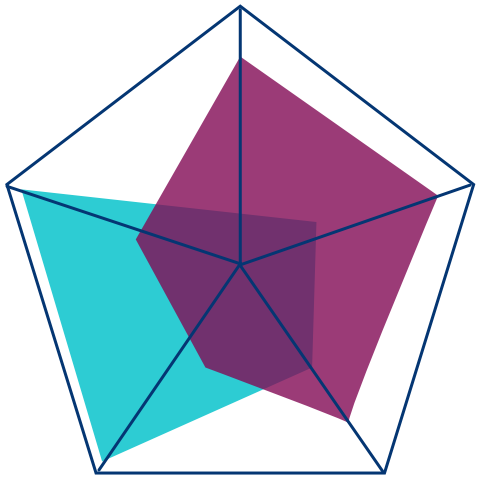
- **The bubble plot:** used to show a variable in three dimensions, position coordinates (x, y) and size.



- **Bubble map**: used to visualize three-dimensional values for geographic regions.

8. Radar charts

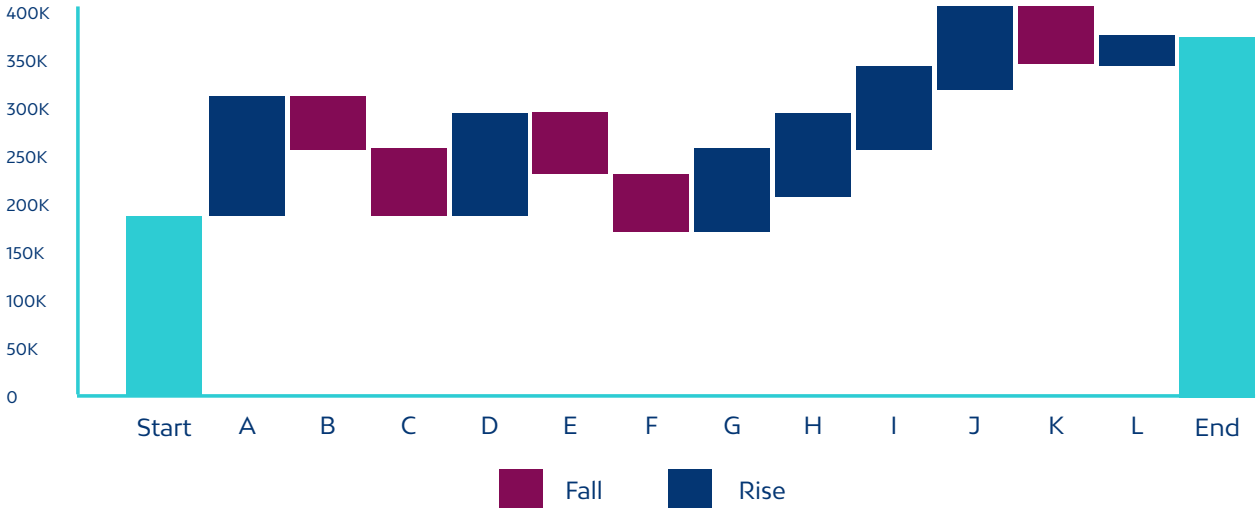
These are **a form of representation built around a regular polygon that is contained within a circle, where the radii that guide the vertices are the axes over which the values are represented.** They are equivalent to graphics with parallel coordinates on polar coordinates. Typically, they are used to represent the behavior of a metric over the course of a set time cycle, such as the hours of the day, months of the year, or days of the week.



Radar chart

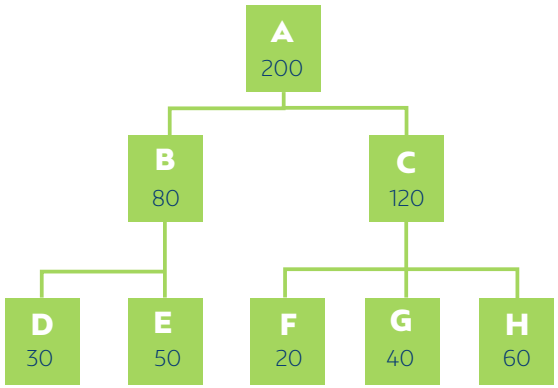
9. Waterfall charts

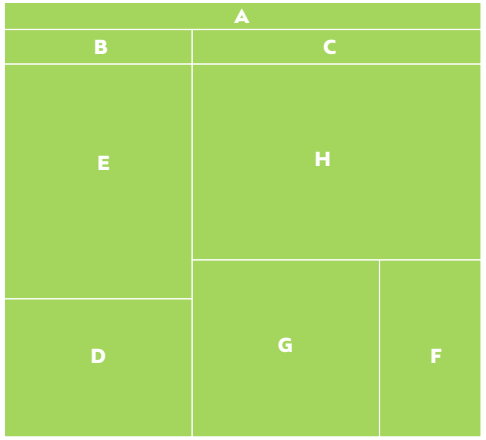
These **help us understand the cumulative effect of positive and negative values on variables in a sequential fashion.**



10. Tree maps

Tree maps **display hierarchical data (in a tree structure) as a set of nested rectangles that occupy surface areas proportional to the value of the variable they represent.** Each tree branch is given a rectangle, which is later placed in a mosaic with smaller rectangles that represent secondary branches. The finished product is an intuitive, dynamic visual of a plane divided into areas that are proportional to hierarchical data, which has been sorted by size and given a color key.

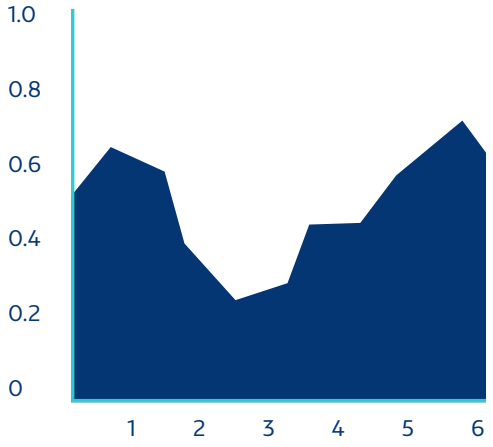




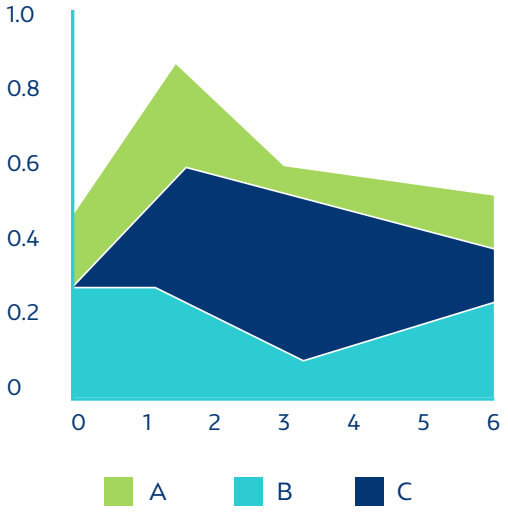
11. Area charts

These **represent the relationship of a series over time, but unlike line charts, they can represent volume.** The three variations on the area chart are:

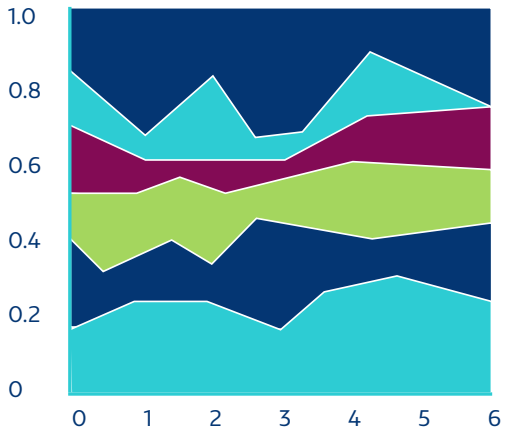
- **Standard area:** used to display or compare a progression over time.
- **Stacked area:** used to visualize relationships as part of the whole, thus demonstrating the contribution of each category to the cumulative total.
- **100% stacked area:** used to communicate the distribution of categories as part of a whole, where the cumulative total does not matter.



Standard area



Stacked area



100% stacked area

Selecting the right graphic to effectively communicate through our visualizations is no easy task. Stephen Few (2009), a specialist in data visualization, proposes taking a practical approach to selecting and using an appropriate graphic:

- **Choose a graphic** that will capture the viewer's attention for sure.
- **Represent the information** in a simple, clear, and precise way (avoid unnecessary flourishes).
- Make it easy to compare data; **highlight trends and differences.**
- **Establish an order** for the elements based on the quantity that they represent; that is, detect maximums and minimums.
- **Give the viewer a clear way to explore the graphic** and understand its goals; make use of guide tags.